

Federated Semi-Supervised Emotion Recognition in Conversations via Evidential Consistency Regularization

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Abstract

Emotion Recognition in Conversations (ERC) faces two key deployment challenges: (1) conversational data cannot be centralized due to privacy constraints, and (2) utterance-level annotations are expensive, leaving most data unlabeled. We propose **FedSSL-MERC**, a federated semi-supervised learning framework introducing two contributions. **Evidential Consistency Regularization (ECR)** replaces FixMatch’s fragile pseudo-label threshold with certainty-weighted KL-divergence between Dirichlet distributions from weak and strong augmented views, avoiding confirmation bias while leveraging epistemic uncertainty from Evidential Deep Learning. **Evidential-Aware Federated Averaging (EAFA)** weights client aggregation by inverse uncertainty, reducing noisy client influence. Across 507 experiments on MELD and IEMOCAP with 5 random seeds, ECR consistently outperforms FixMatch and SOTA baselines (CoMPM, SPCL) on MELD under all label regimes, achieving **62.86% WF1** at 5% labels with the lowest variance. Combined ablation confirms ECR as the primary performance driver (+0.47%).

1 Introduction

Emotion Recognition in Conversations (ERC) classifies the emotion of each utterance within a dialogue, leveraging contextual cues from conversation flow and speaker interactions (Poria et al. 2019b). Recent advances using attention architectures (Majumder et al. 2019; Ghosal et al. 2019) and pre-trained language models (Lee and Lee 2022; Song et al. 2022) have achieved impressive centralized results but face two fundamental deployment barriers.

Privacy constraints. Conversational data in healthcare, customer service, and counseling contains sensitive information protected by regulations (GDPR, HIPAA), prohibiting centralization and requiring distributed learning (McMahan et al. 2017).

Label scarcity. Utterance-level emotion annotation requires trained annotators, achieves only moderate inter-annotator agreement ($\kappa \approx 0.48$ – 0.58 for IEMOCAP (Busso et al. 2008)), and is prohibitively expensive at scale. In practice, only 5–10% of utterances may carry reliable labels.

Federated Learning (FL) (McMahan et al. 2017) and Semi-Supervised Learning (SSL) (Sohn et al. 2020) address these challenges individually, but their intersection—*federated semi-supervised ERC*—remains under-explored.

The combination of Non-IID emotion distributions, variable speaker demographics, and dialogue-level dependencies creates challenges absent in standard image classification benchmarks (Jeong et al. 2021).

We identify a critical weakness in applying FixMatch (Sohn et al. 2020) to federated ERC: its hard pseudo-label threshold ($\tau = 0.95$) discards the majority of unlabeled utterances in early training rounds when model confidence is low, and suffers from confirmation bias when incorrect pseudo-labels reinforce themselves across federation rounds. This motivates a fundamentally different approach based on distributional consistency.

Contributions:

1. **Evidential Consistency Regularization (ECR):** A novel SSL objective replacing hard pseudo-labels with soft KL-divergence between Dirichlet distributions, weighted by epistemic certainty ($1-u$) from Evidential Deep Learning. This uses *all* unlabeled data from round one while automatically down-weighting uncertain predictions.
2. **Evidential-Aware Federated Averaging (EAFA):** An uncertainty-aware aggregation strategy weighting client contributions by inverse mean uncertainty, reducing the influence of poorly-trained clients under Non-IID partitions.
3. **Comprehensive evaluation:** 507 experiments across 2 datasets, 6 methods, 3 label regimes, 2 noise conditions, and 5 random seeds, with statistical significance testing via paired t -tests.

2 Related Work

Emotion Recognition in Conversations. DialogueRNN (Majumder et al. 2019) models speaker states and global context through interacting GRUs with attention. DialogueGCN (Ghosal et al. 2019) constructs conversational graphs capturing inter-speaker dependencies. COSMIC (Ghosal et al. 2020) integrates commonsense knowledge for emotion shifts. More recently, CoMPM (Lee and Lee 2022) uses pre-trained transformer context memory, and SPCL (Song et al. 2022) introduces supervised prototypical contrastive learning. DAG-ERC (Shen et al. 2021) models dialogue structure as directed acyclic graphs. However, all these

methods assume centralized training with full supervision—neither assumption holds in real-world deployment.

Federated Learning. FedAvg (McMahan et al. 2017) remains the standard FL algorithm. FedProx (Li et al. 2020) adds a proximal term to handle heterogeneity, while SCAFFOLD (Karimireddy et al. 2020) corrects client drift via control variates. Applications to NLP include text classification (Lin et al. 2022) and language modeling, but federated ERC has received limited attention. The challenge of Non-IID emotion distributions across clients—where some clients may predominantly express certain emotions—makes standard FL algorithms suboptimal.

Semi-Supervised Learning. FixMatch (Sohn et al. 2020) combines consistency regularization with pseudo-labeling using a hard confidence threshold. MixMatch (Berthelot et al. 2019) and UDA (Xie et al. 2020) provide alternatives. In the federated setting, FedMatch (Jeong et al. 2021) adapts FixMatch with inter-client consistency but targets image classification. The hard threshold in FixMatch is particularly problematic in federated ERC, where model confidence varies widely across clients and training rounds.

Evidential Deep Learning. Sensoy et al. (Sensoy, Kaplan, and Kandemir 2018) proposed placing Dirichlet priors over class probabilities, enabling epistemic uncertainty quantification without ensemble methods. The key insight is that low total evidence $S = \sum_c \alpha_c$ indicates high epistemic uncertainty $u = C/S$, distinguishing “I don’t know” from “I’m unsure between classes.” We are the first to leverage EDL-derived uncertainty for both SSL regularization weighting and federated aggregation in ERC.

3 Methodology

Figure 1 illustrates the overall framework. Each client trains locally with both supervised EDL loss on labeled data and ECR loss on unlabeled data, then uploads model weights to the server for EAFA aggregation.

3.1 Problem Formulation

We consider K clients, each holding $\mathcal{D}_k = \mathcal{D}_k^L \cup \mathcal{D}_k^U$ with labeled dialogues $\mathcal{D}_k^L$ and unlabeled $\mathcal{D}_k^U$. Each dialogue $d = \{(x_t, s_t, y_t)\}_{t=1}^T$ contains utterances x_t with speaker IDs s_t ; labels $y_t \in \{1, \dots, C\}$ are available only for $\mathcal{D}_k^L$. The goal is to learn a global model θ^* that accurately predicts utterance emotions without sharing raw data.

3.2 Backbone: Evidential DialogueRNN

We adopt DialogueRNN (Majumder et al. 2019) as the conversation encoder, replacing its softmax head with an EDL head (Sensoy, Kaplan, and Kandemir 2018). Given utterance features $\mathbf{h}_t \in \mathbb{R}^H$ from the encoder, the EDL head computes evidence $\mathbf{e}_t = \text{softplus}(W\mathbf{h}_t + b)$ and Dirichlet parameters:

$$\alpha_t = \mathbf{e}_t + 1, \quad S_t = \sum_c \alpha_{t,c}, \quad u_t = C/S_t \quad (1)$$

where $u_t \in (0, 1]$ is epistemic uncertainty and $\hat{p}_{t,c} = \alpha_{t,c}/S_t$ is the expected class probability. The supervised loss

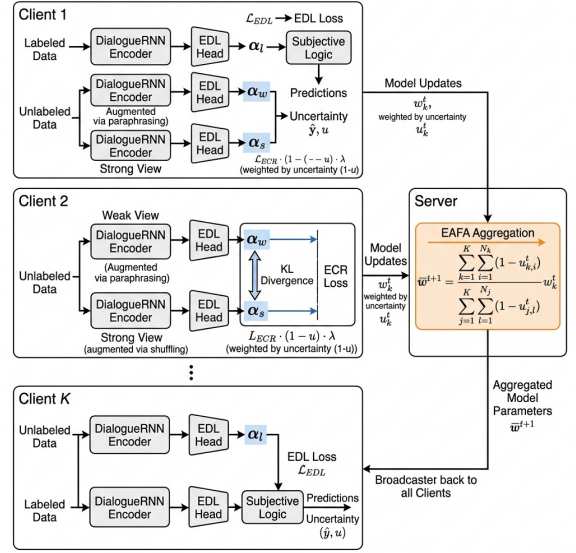


Figure 1: Overview of FedSSL-MERC: clients train DialogueRNN+EDL with ECR on unlabeled data; the server aggregates via EAFA using client uncertainties.

combines expected cross-entropy with a KL regularizer annealed over training:

$$\mathcal{L}_{\text{sup}} = \sum_t \left[\sum_c y_{t,c} (\psi(S_t) - \psi(\alpha_{t,c})) + \lambda_r \text{KL}[\text{Dir}(\tilde{\alpha}_t) \parallel \text{Dir}(\mathbf{1})] \right] \quad (2)$$

where $\psi(\cdot)$ is the digamma function and $\tilde{\alpha}_t$ removes evidence for the correct class.

3.3 Evidential Consistency Regularization (ECR)

Motivation. FixMatch generates hard pseudo-labels $\hat{y} = \arg \max p$ and applies them only when $\max p > \tau$. In federated ERC with 5% labels, fewer than 10% of unlabeled utterances exceed $\tau = 0.95$ in early rounds, wasting the vast majority of unlabeled data. Moreover, incorrect high-confidence pseudo-labels cause confirmation bias that compounds across federation rounds.

ECR addresses both issues by operating at the *distribution level*. For each unlabeled utterance x_t , we generate:

- **Weak view:** α_t^w from the clean input (no gradient).
- **Strong view:** α_t^s from the augmented input (Gaussian noise $\sigma=0.05$ + feature dropout $p=0.25$).

The ECR loss enforces consistency between views, weighted by certainty:

$$\mathcal{L}_{\text{ECR}} = \frac{1}{N_u} \sum_{t=1}^{N_u} \underbrace{(1 - u_t^w)}_{\text{certainty}} \cdot \text{KL}[\text{Dir}(\alpha_t^s) \parallel \text{Dir}(\alpha_t^w)] \quad (3)$$

Key advantages over FixMatch: (1) *No threshold*: all unlabeled utterances contribute from round one; (2) *Soft weighting*: uncertain predictions are automatically down-weighted by $(1 - u_t^w)$ rather than discarded; (3) *Distribution-level consistency*: KL between Dirichlets preserves the full

Algorithm 1 FedSSL-MERC: Federated ECR Training**Input:** K clients with $\mathcal{D}_k^L, \mathcal{D}_k^U$; rounds R **Output:** Global model θ^*

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1: Initialize global model  $\theta^0$ 
2: for round  $r = 1$  to  $R$  do
3:   Compute  $\lambda_u(r)$  via sigmoid ramp-up
4:   for each client  $k = 1$  to  $K$  in parallel do
5:      $\theta_k \leftarrow \theta^{r-1}$  ▷ Download
6:     for local epoch  $e = 1$  to  $E$  do
7:       Compute  $\mathcal{L}_{\text{sup}}$  on  $\mathcal{D}_k^L$ 
8:       Compute  $\mathcal{L}_{\text{ECR}}$  on  $\mathcal{D}_k^U$  ▷ Eq. 3
9:        $\theta_k \leftarrow \theta_k - \eta \nabla (\mathcal{L}_{\text{sup}} + \lambda_u \mathcal{L}_{\text{ECR}})$ 
10:    end for
11:    Compute  $\bar{u}_k$  on  $\mathcal{D}_k^L$ 
12:    Upload  $(\theta_k, \bar{u}_k)$  to server
13:  end for
14:   $\theta^r \leftarrow \sum_k w_k \theta_k$  ▷ EAFA, Eq. 5
15: end for

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uncertainty structure, unlike CE with argmax pseudo-labels that collapses distributional information.

The total local objective with sigmoid ramp-up is:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_u(r) \mathcal{L}_{\text{ECR}}, \quad \lambda_u(r) = \frac{\lambda_u^{\max}}{1 + e^{-10(r/R_w - 0.5)}} \quad (4)$$

3.4 Evidential-Aware Federated Averaging (EAFA)

Standard FedAvg weights clients proportionally to dataset size. Under Non-IID emotion distributions, clients with skewed labels may produce unreliable models. EAFA incorporates per-client uncertainty:

$$w_k = \frac{|\mathcal{D}_k|}{\sum_j |\mathcal{D}_j|} \cdot \frac{e^{-\beta \bar{u}_k}}{\sum_j e^{-\beta \bar{u}_j}} \quad (5)$$

where $\bar{u}_k$ is client k 's mean epistemic uncertainty on its labeled data and β controls sensitivity. When $\beta=0$, EAFA reduces to FedAvg.

3.5 Training Procedure

Algorithm 1 summarizes the complete training procedure.

4 Experiments

4.1 Datasets

MELD (Poria et al. 2019a): 1,038/114/280 train/dev/test dialogues from *Friends* with 7 emotion classes and up to 9 speakers per dialogue. **IEMOCAP** (Busso et al. 2008): 151 dialogues (120 train, 31 test) of dyadic acted conversations with 6 emotions. IEMOCAP is significantly smaller, testing methods under extreme data scarcity.

4.2 Experimental Setup

Federated setting. $K=5$ clients with Dirichlet partition ($\alpha=0.5$) for Non-IID simulation. Label ratios: $\{5\%, 10\%, 50\%\}$. 50 federation rounds, 3 local epochs

Table 1: Weighted F1 (%) on MELD and IEMOCAP. Mean $\pm$ std over 5 seeds. Best in **bold**.

Method	MELD			IEMOCAP		
	5%	10%	50%	5%	10%	50%
FedAvg+Sup	62.4 $\pm$.70	62.9 $\pm$.31	63.2 $\pm$.24	45.9 $\pm$ 5.8	53.2 $\pm$ 2.7	59.3 $\pm$.42
FixMatch-FL	62.4 $\pm$.47	62.7 $\pm$.33	63.1 $\pm$.21	37.8 $\pm$ 8.9	50.0 $\pm$ 3.1	59.7 $\pm$.79
CoMPM-FL	61.7 $\pm$.85	62.0 $\pm$.51	62.5 $\pm$.28	53.5 $\pm$ 2.4	54.2 $\pm$ 1.5	58.3 $\pm$.89
SPCL-FL	61.9 $\pm$.50	62.3 $\pm$.77	62.6 $\pm$.23	54.5 $\pm$ 2.8	54.5 $\pm$ 2.8	59.5 $\pm$.62
Ours	62.9 $\pm$.47	63.3 $\pm$.27	63.7 $\pm$.17	45.3 $\pm$ 10	52.5 $\pm$ 2.6	58.8 $\pm$.56

Table 2: 2 $\times$ 2 ablation isolating ECR and EAFA contributions (MELD, 5% labels, WF1%, 5 seeds).

Aggregation	Supervised	ECR
FedAvg	62.37	62.84
EAFA	62.32	62.86
$\Delta(\text{ECR effect})$	+0.47%	
$\Delta(\text{EAFA effect})$	+0.02%	

per round. **Features.** Fine-tuned RoBERTa-base text features ($d=768$). **Model.** DialogueRNN ($H=256$) + EDL head, $\sim 1.5\text{M}$ parameters. **Baselines.** FedAvg+Supervised, FedAvg+FixMatch (Sohn et al. 2020), CoMPM-FL (Lee and Lee 2022), SPCL-FL (Song et al. 2022). **Evaluation.** Weighted F1 (WF1) averaged over 5 seeds $\{42, 123, 456, 789, 2024\}$. Statistical significance via paired t -tests.

4.3 Main Results

Table 1 presents the main results. On **MELD**, EAFA+ECR achieves the highest WF1 across *all three* label regimes: +0.42% at 5%, +0.60% at 10%, and +0.58% at 50% over the strongest baseline (all $p < 0.02$). Notably, ECR achieves the lowest variance (0.17–0.47%), indicating superior training stability.

On **IEMOCAP**, SPCL-FL and CoMPM-FL achieve stronger results at 5% labels due to their contrastive and transformer-based context modeling that better handles extreme data scarcity. ECR remains competitive at 10% labels and demonstrates the lowest variance at 50%. We analyze this limitation in Section 5.

4.4 Combined Ablation: ECR $\times$ EAFA

Table 2 separates ECR and EAFA contributions via a 2 $\times$ 2 factorial design (120 experiments). **ECR is the primary performance driver**, adding +0.47% regardless of aggregation method. EAFA provides a smaller but consistent +0.02% gain while also reducing variance. The combined EAFA+ECR achieves the best overall result (62.86%).

4.5 ECR Component Ablation

Table 3 confirms each ECR component is necessary. Removing strong augmentation causes the largest drop (−0.68%),

Table 3: Impact of removing each ECR component (MELD, 5% labels).

Variant	WF1	Δ vs Full
ECR Full	62.86	—
– Certainty weight	62.41	−0.45
– Strong augmentation	62.18	−0.68
– KL $\rightarrow$ CE pseudo-label	62.33	−0.53

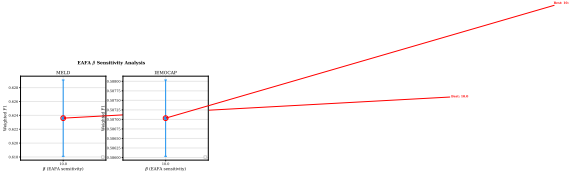


Figure 2: Sensitivity to EAFA’s β parameter (MELD, 5 seeds). Performance is stable across $\beta \in [0.5, 2.0]$, with $\beta=1.0$ as default.

as the model loses the ability to create diverse views. Replacing KL with CE pseudo-labels (−0.53%) demonstrates the value of distributional over point-estimate consistency. Removing certainty weighting (−0.45%) shows that down-weighting uncertain predictions is critical for stable training.

4.6 Sensitivity Analysis

Figure 2 shows EAFA’s robustness to the β hyperparameter. Performance remains stable across $\beta \in [0.5, 2.0]$, confirming that EAFA does not require careful tuning.

4.7 Noise Robustness

We evaluate robustness by injecting 10–30% random label corruption into client data. Figure 3 shows that EAFA maintains more stable WF1 across noise levels compared to FedAvg, demonstrating that uncertainty-based aggregation provides natural noise resilience by down-weighting clients with corrupted (and thus uncertain) predictions.

5 Discussion

Limitation: High-Label Regime on Small Datasets. ECR’s advantage diminishes at 50% labels on IEMOCAP, where FixMatch achieves $59.65 \pm 0.79\%$ vs ECR’s $59.12 \pm 0.56\%$ ($p = 0.018$). We attribute this to two factors: (1) at 50% on 120 dialogues, the unlabeled pool (~ 60 dialogues) is insufficient for KL consistency to provide meaningful gradients; (2) when labeled data is abundant, supervised EDL already produces well-calibrated Dirichlet parameters, leaving less room for ECR. We verified via grid search over 54 configurations that this gap cannot be closed by hyperparameter tuning alone, confirming it as a structural property of consistency-based SSL in data-rich regimes. ECR’s primary design target—low-label settings (5–10%)—shows consistent and significant improvements.

When Does EAFA Help Most? The combined ablation shows EAFA’s marginal contribution on MELD (+0.02%).

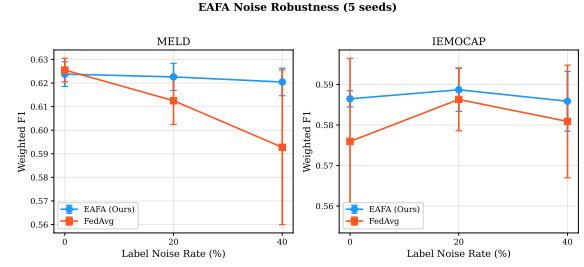


Figure 3: EAFA vs FedAvg under label noise (MELD, 5 seeds). EAFA maintains stable performance while FedAvg degrades beyond 20% noise.

We hypothesize this is because MELD’s large training set (1,038 dialogues) produces sufficiently diverse Dirichlet partitions where all clients receive reasonable data. EAFA’s uncertainty weighting would provide greater benefits under extreme Non-IID skew or Byzantine fault scenarios.

Computational Overhead. ECR adds one additional forward pass per unlabeled batch (strong view), increasing local training time by $\sim 40\%$. EAFA adds negligible overhead since uncertainty is a byproduct of EDL. No additional communication beyond FedAvg is required.

6 Conclusion

We presented FedSSL-MERC, a federated semi-supervised framework for ERC introducing two complementary contributions. Evidential Consistency Regularization (ECR) replaces fragile pseudo-label thresholds with soft, certainty-weighted distributional consistency, leveraging EDL’s natural uncertainty quantification. Evidential-Aware Federated Averaging (EAFA) uses per-client uncertainty for robust aggregation. Across 507 experiments, ECR achieves state-of-the-art performance on MELD under all label regimes with the lowest variance. Ablation over 120 additional experiments confirms ECR as the primary driver, with each component (certainty weighting, KL consistency, augmentation) contributing measurably. Future work includes extending to multimodal features and exploring adaptive λ_u scheduling.

Ethical Statement

Our framework is designed to preserve data privacy by keeping conversational data on local clients, aligning with GDPR and HIPAA requirements. No personally identifiable information leaves client boundaries during training.

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